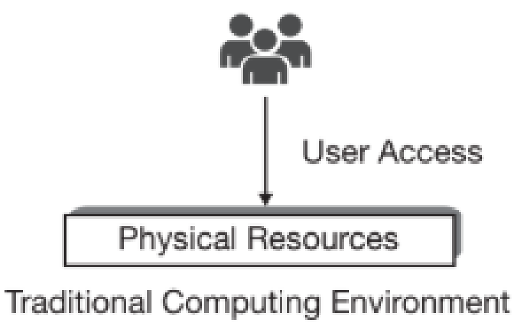
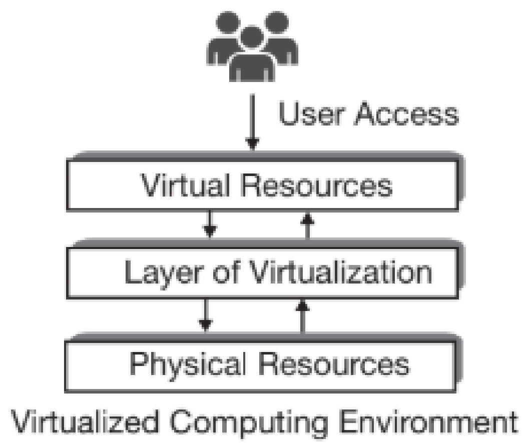
Cloud Computing
Module 1: Understanding Virtualization

Cloud Computing:
It is a technology that enables the delivery of various computing services over the internet, allowing users to access and store data remotely instead of relying on local servers or personal computers.
- Basically: storing and accessing data and programs over the internet rather than the computer's hard disks.

Virtualization:
Virtualization refers to the representation of physical computing resources in simulated form having been made through the software. This special layer of software (installed over active physical machines) is referred to as the layer of virtualization. This layer transforms the physical computing resources into virtual form which users use to satisfy their computing needs.





Virtualization provides a level of logical abstraction that liberates user-installed software (starting from operating system and other systems as well as application software) from being tied to a specific set of hardware. Rather, the users install everything over the logical operating environment (rather than physical ones) having been created through virtualization.

Features of Virtualization:
- Partitioning: Multiple virtual servers can run on a physical server at the same time.
- Encapsulation of data: All data on the virtual server is encapsulated in a file format.
- Isolation: The server running on the physical server is safely separated and doesn't affect each other.
- Hardware Independence: When the virtual server runs, it can migrate to a different platform

Types of Virtualization:
- Operating systems
- Storage
- Memory
- CPU
- Hardware

Operating System Virtualization

Definition: Operating System virtualization allows multiple operating systems to run concurrently on a single physical machine. This is achieved through a hypervisor that manages the virtual machines (VMs).

Features:

- Multi-tenancy: Multiple users can share the same physical resources without interference.
- Isolation: Each VM operates independently, enhancing security.
- Resource Allocation: Dynamic allocation of resources based on demand.

Advantages:

- Resource Optimization: Better utilization of hardware by running multiple VMs on a single server.
- Cost Reduction: Decreases hardware costs and maintenance expenses.
- Scalability and Flexibility: Easy to scale resources up or down as needed.
- Disaster Recovery: Simplifies backup and recovery processes through VM snapshots.

Disadvantages:

- Complexity in Management: Requires skilled personnel to manage the virtualization environment.
- Performance Overhead: Some performance degradation can occur due to the virtualization layer.
- Security Risks: Vulnerabilities in the hypervisor can expose all VMs to potential threats

Storage Virtualization

Definition: Storage virtualization abstracts multiple physical storage devices into a single, unified storage pool, simplifying data management.

Features:

- Dynamic Allocation: Storage can be allocated or reallocated based on demand.
- Unified Management Interface: Provides a consistent interface for managing diverse storage types.

Advantages:

- Improved Resource Utilization: Optimizes storage use by pooling resources.
- Cost Efficiency: Reduces hardware costs by minimizing the need for additional storage devices.
- Enhanced Disaster Recovery: Simplifies data migration and backup processes.

Disadvantages:

- Initial Complexity: Setting up storage virtualization can be complex and time-consuming.
- Dependency on Software: Performance may be impacted by the limitations of the virtualization software.
- Potential Security Risks: Centralized control may introduce vulnerabilities

Memory Virtualization

Definition: Memory virtualization allows a computer to use secondary storage as if it were part of its main memory (RAM), effectively increasing available memory.

Features:

- Paging and Segmentation: Utilizes techniques like paging to manage memory allocation dynamically.
- Swapping Mechanism: Transfers inactive data from RAM to disk storage when necessary.

Advantages:

- Increased Memory Availability: Allows more applications to run simultaneously than would be possible with physical RAM alone.
- Cost-effective Memory Management: Reduces the need for additional physical memory by optimizing usage.

Disadvantages:

- Performance Overhead: Accessing data from disk is slower than accessing RAM, potentially leading to performance bottlenecks.
- Complexity in Management: Requires careful configuration to optimize performance

CPU Virtualization

Definition: CPU virtualization allows multiple virtual processors to run on a single physical processor by abstracting CPU resources.

Features:

- Time-Slicing: Allocates CPU time among VMs through time-sharing techniques.
- Virtual CPUs (vCPUs): Each VM is assigned one or more vCPUs that correspond to physical CPU resources.

Advantages:

- Enhanced Resource Utilization: Maximizes CPU usage by running multiple workloads simultaneously.
- Isolation and Security: Each VM operates independently, minimizing risks associated with shared resources.

Disadvantages:

- Performance Limitations: Heavy workloads may lead to contention for CPU resources, impacting performance.
- Management Complexity: Requires sophisticated management tools to monitor and allocate CPU resources effectively.

Hardware Virtualization

Definition: Hardware virtualization involves creating virtual versions of hardware components, allowing multiple operating systems to run on a single physical machine.

Features:

- Hypervisors (Type 1 and Type 2): Type 1 runs directly on hardware; Type 2 runs on an existing OS.
- Direct Access to Hardware Resources: VMs can access hardware components directly through the hypervisor.

Advantages:

- Improved Efficiency and Performance: Direct access can enhance performance compared to OS-level virtualization.
- Flexibility in Deployment: Supports various operating systems and applications on a single platform.

Disadvantages:

- Higher Resource Requirements: May require more powerful hardware to support multiple VMs efficiently.
- Complex Setup and Maintenance: Initial setup can be complex, requiring specialized knowledge

Understanding Abstraction:

Abstraction is the process of hiding the complex and non-essential characteristics of a system. Through abstraction, a system can be presented in a simplified manner for some particular use after omitting unwanted details from users. In computing, abstraction is implemented through the layers of software. The layer of the operating system can be treated as a layer of abstraction.

In cloud computing, resource virtualization, which adds a layer of software over physical computing resources to create virtual resources, acts as a layer of abstraction. This abstraction makes it easier to offer more flexible, reliable and powerful service. Virtualization creates a layer of abstraction and masks physical resources from external access.

Hypervisor:

A hypervisor, also known as a virtual machine monitor (VMM) or virtualizer, is a software, firmware, or hardware that enables multiple virtual machines (VMs) to run on a single physical machine. Each VM has its own operating system and applications7. The hypervisor allocates physical computing resources, such as processing power, memory, and storage, to each VM as needed.

How Hypervisors Work:

Hypervisors pool and reallocate computing resources among VMs, making virtualization possible3.

They abstract and isolate VMs from the underlying physical server hardware, enabling efficient resource use, simpler maintenance, and reduced costs.

The hypervisor manages the scheduling of VM resources against the physical resources. The physical hardware executes instructions requested by the VMs, while the hypervisor manages the schedule. VMs communicate with the hypervisor through application programming interface (API) calls.

Types of Hypervisors:

- Type 1: Also known as bare-metal or native hypervisors, these run directly on the host's hardware, managing guest operating systems. They take the place of a host operating system, scheduling VM resources directly to the hardware. Examples include KVM, Microsoft Hyper-V, and VMware vSphere.
- Type 2: Also known as hosted hypervisors, these run on a conventional operating system as a software layer or application. They abstract guest operating systems from the host operating system, scheduling VM resources against a host operating system, which is then executed against the hardware. VMware Workstation and Oracle VirtualBox are examples of type 2 hypervisors and are better for individual users.

Benefits of Hypervisors:

- Efficiency VMs can be created instantly, providing higher accuracy and efficiency in allocating resources for complex workloads.
- Adaptability Type 1 hypervisors allow guest operating systems and their applications to run on multiple different hardware types because the hypervisor separates each guest OS from the host computer’s hardware.
- Agility Hypervisors provide agile utilization of physical servers and a cost-efficient way to run multiple VMs, rather than using multiple servers to complete the same task.

Hypervisor-based virtualization techniques: Hypervisor-based virtualization techniques can be divided into three categories as full virtualization, para-virtualization and hardware-assisted virtualization. The techniques being used in these three approaches are entirely different from one another.

Full Virtualization

In full virtualization (also called as native virtualization), the hypervisor fully simulates or emulates the underlying hardware. Virtual machines run over these virtual sets of hardware. The guest operating systems assume that they are running on actual physical resources and thus remain unaware that they have been virtualized. This enables the unmodified versions of available operating systems (like Windows, Linux and others) to run as guest OS over hypervisor. In this model, it is the responsibility of the hypervisor to handle all OS-to-hardware (i.e. guest OS to physical hardware) requests during running of guest machines. The guest OS remains completely isolated from physical resource layers by the hypervisor. This provides flexibility as almost all of the available operating systems can work as guest OS. VMware's virtualization product VMWare ESXi Server and Microsoft Virtual Server are few examples of full virtualization solutions. In full virtualization technique, the guest operating systems can directly run over the hypervisor.

Para-Virtualization or OS-Assisted Virtualization

'Para' is an English affix of Greek origin that means 'beside' or 'alongside'. In full virtualization, the operating systems which run in VMs as guest OSs need not have any prior knowledge that they will run over a virtualized platform. They do not need any special modification (or, functionality incorporation) for running over the hypervisors and are installed in their original form. The whole virtualization activities are managed by the hypervisor like translating instructions and establishing communication between different guest OSs and the underlying hardware platform. In para-virtualization, a portion of the virtualization management task is transferred (from the hypervisor) towards the guest operating systems. Normal versions of available operating systems are not capable of doing this. They need special modification for this capability inclusion. This modification is called porting. Each guest OS is explicitly ported for the para-application program interface (API). Thus in para-virtualization, each guest OS needs to have prior knowledge that it will run over the virtualized platform. Moreover, it also has to know on which particular hypervisor they will have to run. Depending on the hypervisor, the guest OS is modified as required to participate in the virtualization management task.The unmodified versions of available operating systems (like Windows, Linux) cannot be used in para-virtualization. Since it involves modifications of the OS the paravirtualization is sometimes referred to as OS-Assisted Virtualization too. This technique relieves the hypervisor from handling the entire virtualization tasks to some extent. Best known example of para-virtualization hypervisor is the open-source Xen project which uses a customized Linux kernel. Para-virtualization requires hypervisor-specific modifications of guest operating systems.

<u>Advantages</u>

- Para-virtualization allows calls from guest OS to directly communicate with hypervisor (without any binary translation of instructions). The use of modified OS reduces the virtualization overhead of the hypervisor as compared to the full virtualization.
- In para-virtualization, the system is not restricted by the device drivers provided by the virtualization software layer. In fact, in para-virtualization, the virtualization layer (hypervisor) does not contain any device drivers at all. Instead, the guest operating systems contain the required device drivers.

<u>Limitations</u>

- Unmodified versions of available operating systems (like Windows or Linux) are not compatible with para-virtualization hypervisors. Modifications are possible in Open- source operating systems (like Linux) by the user. But for proprietary operating systems (like Windows), it depends upon the owner. If the owner agrees to supply the required modified version of the OS for a hypervisor, then only that OS becomes available for the paravirtualization system.
- Security is compromised in this approach as the guest OS has a comparatively more control of the underlying hardware. Hence, the users of some VM with wrong intentions have more chances of causing harm to the physical machine. Para-virtualization can provide enhanced virtualization performance at the cost of security.

Benefits of virtualizing physical computing resources:

- Efficient hardware use With virtualization, companies can create digital servers, or VMs, on a single physical server and specify the operating system requirements for the VMs and use them like the physical servers.
- Cost savings Improved hardware utilization can mean savings on additional physical resources, as well as reducing the need for power, space, and cooling in the data center.
- Isolated environments Because they're separated from the rest of a system, VMs won't interfere with what's running on the host hardware, and they are a good option for testing new applications or setting up a production environment.
- Faster application migration Because VM configurations are defined by software, VMs can be quickly created, removed, cloned, and migrated.
- Server consolidation By virtualizing servers, many virtual servers can be placed on each physical server to improve hardware utilization. Server consolidation leads to improved resource

utilization when resources are allocated to where they are needed because a host machine can be divided into multiple VMs.
- Disaster recovery VMs provide additional disaster recovery options by enabling failover that could previously only be achieved through additional hardware.

Business Benefits:
- Radically improves the flexibility and availability of computing resources
- Organizations can gain in business lower H/W cost
- Improvement in server utilization
- Faster provisioning of applications and resources.
- Faster and easier backup and recovery of key application workloads and data
- Minimized or eliminated downtime
- Increased IT productivity, efficiency, agility and responsiveness.

Emulation: Emulation in computing is done by making one system imitating another. This means a system having some architecture is made to support an instruction set of some other machine architecture. For example, let a piece of software be made for architecture ‘A’ and is not supported by architecture ‘B’. Through emulation, it is possible to imitate the working of system ‘A’ on system ‘B’ (i.e. architecture ‘B’) and then the piece of software to run on system B. Emulators can be software or hardware both. Emulation software converts binary data written for execution on one machine to an equivalent binary form suitable to execute on another machine. This is done by translating the binary instructions. There are two ways for implementation of emulations like interpretation and binary translation.
- In binary translation (also known as recompilation), a total conversion of the binary data (made for the emulated platform) is done. The conversion recompiles the whole instruction into another binary form suitable to run on the actual or targeted platform. There are two types of binary translation like static recompilation and dynamic recompilation.
- In interpretation, each instruction is interpreted by the emulator every time it is being encountered. This method is easier to implement but slower than the binary translation process. The principle of emulation in computing is instruction-set translation through interpretation.

Top Virtualization Security Issues:
- External Attacks: Cybercriminals seek vulnerabilities to gain unauthorized access to networks and systems5. Virtual environments can be more vulnerable due to their complexity and interconnected nature. An external attack on a virtualized resource can result in data breaches, service disruptions, and reputational damage5. Robust security measures, strong access controls, encryption, and regular updates are essential to protect against these threats5. If attackers gain access to your host-level or VMware vCenter server, this opens doors for them to access other important VMs, or even create a user account with admin rights that could be used over a long period of time to collect or destroy sensitive company data.
- VM Escape: This is a critical security vulnerability where an attacker breaks out of a VM and interacts directly with the hypervisor or other VMs on the same host. This breach can compromise the entire host system, leading to potential access of other VMs and their data5.

Strong isolation between VMs and regular updates and patches for both the hypervisor and the VMs are crucial to mitigate this risk.

- Hypervisor Attacks: Attacks targeting the hypervisor, which has complete control over VMs, can have severe consequences, including the compromise of all VMs hosted on the hypervisor5. These attacks may exploit vulnerabilities in the hypervisor software itself or leverage misconfigurations. Defending against hypervisor attacks involves keeping the hypervisor software up-to-date with the latest security patches and implementing a minimal attack surface on the hypervisor by disabling unnecessary functions or services can reduce vulnerabilities.
- VM Sprawl: VM sprawl is a situation where the number of VMs in an environment grows to the point where they become difficult to manage and secure5. This can occur when VMs are created without proper planning and management, or used to test new applications, features or tools. Each of these abandoned virtual machines does not receive software and security updates, and may not have other security measures like authentication, which can make them an entry point for a cyberattack. Implementing policies for periodic review and decommissioning of unused VMs is crucial.
- Malware: Virtual machines are susceptible to viruses, malware, and ransomware attacks26. Many virtualized resources run on standard operating systems and are susceptible to the same malware threats as traditional systems5. Robust anti-malware measures, including anti-malware solutions, regularly updating and patching systems, and educating users about the risks of malware are essential.
- Network Configuration: Making poor configuration choices, like allowing file sharing between VMs, or leaving unused firewall ports open could be all that's needed for a hacker to gain access to your virtual infrastructure.
- Access Controls: Unauthorized access to virtual infrastructure poses a significant threat. Strong access controls, including secure passwords, multi-factor authentication, and role-based access, are vital.
- Malicious virtual appliances: Malicious virtual appliances would be a threat in public or private cloud environments1. Since you install/use these appliances, there's an element of trust you have given them. The malicious system would then attempt to find vulnerabilities through its "trust" and exploit them.
- Data leakage through offline images: Offline backups are essential for disaster recovery, but outdated security configurations of offline VMs pose risks when brought back online6. Attackers or malicious insiders could collect valuable data from snapshots3. Snapshots are meant to be retained for only a short time.

High-level language virtual machine (HLL VM)

A high-level language virtual machine (HLL VM) is designed to provide platform independence when implementing high-level programming languages45. High-level languages are designed to be easier for humans to understand and write. They use more abstract commands that are less dependent on the specifics of the underlying computer hardware.

<u>How HLL VMs work:</u>

- They offer a platform-independent programming environment that abstracts away details of the underlying hardware or operating system, allowing a program to execute the same way on any platform.

- HLL VMs offer a high-level abstraction, specifically that of a high-level programming language.
- They are implemented using an interpreter, and performance comparable to compiled programming languages can be achieved through just-in-time compilation.

Machine or Server-level virtualization
Machine virtualization (also called server virtualization) is the concept of creating a virtual machine (or virtual computer) on an actual physical machine. The parent system on which the virtual machines run is called the host system, and the virtual machines are themselves referred to as guest systems. In conventional computing systems, there has always been a one-to-one relationship between physical computer and operating system. At a time, a single OS can run over them. Hardware virtualization eliminates this limitation of having a one-to-one relationship between physical hardware and operating system. It facilitates the running of multiple computer systems having their own operating systems on a single physical machine. The OS of the guest systems running over the same physical machine need not to be similar. All these virtual machines running over a single host system, remain independent of each other. Operating systems are installed into those virtual machines. These guest systems can access the hardware of the host system and can run applications within its own operating environment.

Resource Pooling: Resource pooling in cloud computing refers to the practice of combining and managing various computing resources (like processing power, storage, and network bandwidth) in a centralized way, which is dynamically allocated to multiple users or clients based on demand. This allows for greater efficiency, scalability, and flexibility in resource utilization.

Key Aspects

- **Virtualization**: Cloud providers use virtualization technologies to create virtual instances of physical resources (like CPUs, memory, and storage) to serve multiple clients. This abstraction allows multiple virtual machines (VMs) to run on a single physical server.
- **Shared Resources:**Resources like servers, storage devices, and networking equipment are shared among multiple users or tenants. The physical resources are not dedicated to a single user, but are instead distributed based on demand.
- **Dynamic Allocation:** Resources can be allocated and reallocated dynamically. For example, if one user needs more storage or processing power, the system can allocate additional resources on-the-fly, without the user needing to request or manage them manually.
- **Multi-Tenancy:** Multiple clients (tenants) can share the same physical resources while remaining logically separated from each other. This is achieved through techniques like virtualization and containerization.
- **Efficiency and Cost Savings:** Cloud providers can optimize resource usage and reduce waste. Since resources are pooled, they can be distributed more effectively across different customers, improving efficiency and reducing costs.

Benefits of Resource Pooling in Cloud Computing:

- **Scalability:** Cloud providers can scale resources up or down based on demand. For instance, during periods of high traffic, additional resources can be temporarily allocated to meet the increased load.
- **Cost Efficiency:** Pooling allows providers to maximize the utilization of resources, leading to better cost distribution across users. Customers only pay for the resources they use.
- **Flexibility**: Clients can choose from a wide range of resource configurations and change their usage depending on their needs.
- **Load Balancing**: Cloud services can distribute workloads evenly across different resources in the pool, preventing overloading of a single resource and ensuring reliability and uptime.

Example:

Imagine a cloud service provider with a data center containing hundreds of physical servers. Rather than each customer having their own dedicated physical server, the provider pools the resources (CPU, memory, storage) from all those servers and dynamically allocates them to customers based on their needs. If one customer requires more storage for a brief period, that customer can borrow additional space from the shared pool without impacting other users.

Resource Sharing:

Resource sharing in cloud computing refers to the ability to allocate and distribute computing resources like storage, processing power, and networking capabilities among multiple users or applications. This allows for efficient utilization of resources, enhanced scalability, and reduced costs.

Key Concepts:

- Resource Pools: Aggregated resources form a resource pool to handle dynamic IT demands.
- Dynamic Provisioning: Resources are allocated on-demand, adapting to changing loads, rather than being permanently assigned.

- Multi-Tenancy: Multiple customers share the same infrastructure while maintaining data isolation. This is a key feature in public clouds.
- Centralized Management: Resources are managed from centralized pools, optimizing distribution.
- Location Independence: Users typically don't have control over the physical location of resources.

<u>Benefits of Resource Sharing:</u>

- Cost Reduction: Centralized resource creation and sharing reduces the need to repeatedly create resources in each account.
- Scalability: Resources can be scaled up or down based on needs without major infrastructure changes.
- Improved Resource Utilization: Distributing resources among applications can increase the average utilization of assets.
- Centralized Management: Operations, maintenance, and security policies can be managed in a centralized manner.

<u>Resource Sharing Implementation:</u>

- Virtualization: Used to create a unified hardware resource pool for management.
- Tenancy Models:
    - Single Tenancy: Dedicated instance of an application and infrastructure for each customer, offering high security but higher costs.
    - Multi-Tenancy: Multiple customers share a single instance of an application and infrastructure, reducing costs but requiring logical separation to maintain data security.
- Resource Sharing Service: Resources within one account can be directly shared with another account. Resources can also be shared based on resource directories.

Cloud Resource provisioning:

Cloud resource provisioning is the process of allocating and configuring IT resources such as servers, storage, databases, processing power, and networking to meet specific application requirements. This ensures efficiency and seamless business operations by providing the right resources at the right time.

Types of Cloud Resource Provisioning:

- Advanced Provisioning: A formal service contract is signed with the cloud provider, who prepares and delivers the agreed-upon resources or services. The customer is charged a flat fee or billed monthly.
- Dynamic Provisioning: Cloud resources are deployed to match a customer's fluctuating demands, scaling up during usage spikes and down when demands decrease. Billing is on a pay-per-use basis. When used to create a hybrid cloud environment, it's referred to as cloud bursting.
- Manual Provisioning: IT administrators handle the allocation and configuration, offering a high level of control but less adaptability to dynamic workload changes. It suits static workloads with predictable resource demands.
- Automated Provisioning: Scripts or tools minimize human intervention, speeding up deployment and enhancing responsiveness to evolving demands. Ideal for environments with varying workloads.
- Cost-Based Provisioning: Focuses on minimizing the cost of running cloud services by dynamically adjusting resource allocation based on pricing models, provisioning or de-provisioning VMs based on budget constraints.